Homework 4

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1. Let G be a group, and let H, K be subgroups of G. Let $HK = \{hk : h \in H, k \in K\}$. Prove that HK is a subgroup of G if and only if $HK = KH$. Conclude that if H is normal, then HK is always a subgroup. Also show that if HK is a subgroup, it is the smallest subgroup containing H and K.

Proof. If HK is a subgroup of G, then given $g_1 = hk$ for some $h \in H$ and $k \in K$,

$$g_1^{-1} = (hk)^{-1} = k^{-1}h^{-1} \in KH.$$

Since g_1^{-1} is also in HK, the same argument gives us $(g_1^{-1})^{-1} = g_1 \in KH$. Likewise, let $g_2 = kh$ $(h \in H, k \in K)$. Then $g_2^{-1} = h^{-1}k^{-1} \in HK$, and since HK is closed under inverses, $g_2 \in HK$.

On the other hand, if $HK = KH$, then given $h_1, h_2 \in H$ and $k_1, k_2 \in K$, we have

$$(h_1k_1)(h_2k_2)^{-1} = (h_1k_1)(k_2^{-1}h_2^{-1}) \in HK$$

since both terms on the right hand side are in $HK = KH$.

Indeed, if H is normal, then given $h_1 \in H$ and $k \in K$, there exists $h_2 \in H$ such that

$$\begin{aligned} kh_1k^{-1} &= h_2 \\ \implies kh_1 &= h_2k \in HK \\ \implies h_1k^{-1} &= k^{-1}h_1 \in KH, \end{aligned}$$

so $HK = KH$ and thus HK is a subgroup.

Also, for any subgroup J containing H and K, given any $h \in H$ and $k \in K$, h and k are also in J, so $hk \in J$, and thus $HK \subset J$. □

2.

(a) Let G be a group. Prove that if $G/Z(G)$ is cyclic then G is abelian.

(b) Suppose G is a group of order pq for primes p and q. Prove that either G is abelian or has trivial center.

Proof.

(a) UNFINISHED Let $g, h \in G$, and choose $x \in G$ such that $xZ(G)$ generates $G/Z(G)$. If either g or h is in $Z(G)$, then $gh = hg$ by definition, so consider when neither is in $Z(G)$. Then $gZ(G) = x^m Z(G)$ and $hZ(G) = x^n Z(G)$ for some $m, n \in \mathbb{Z}^+$, so (letting $\bar{\cdot}$ denote $\cdot Z(G)$),

$$\bar{g}\bar{h} = \overline{x^m}\overline{x^n} = \overline{x^{m+n}} = \overline{x^{n+m}} = \overline{x^n}\overline{x^m} = \bar{h}\bar{g}$$

(b) By Lagrange's theorem, $|Z(G)| \in \{1, p, q, pq\}$. If $|Z(G)| = p$ (resp. q), then $|G/Z(G)| = q$ (resp. p), so $G/Z(G)$ is cyclic since it has prime order and part (a) shows that G is abelian. If $|Z(G)| = pq$, then $Z(G) = G$, so G is abelian. Otherwise, $|Z(G)| = 1$, so $Z(G)$ is trivial.

□

3. Let G be a finite group, and suppose that H is a subgroup of index p where p is the smallest prime that divides $|G|$. Prove that H is normal in G.

Proof. NOT STARTED □

4. Suppose $\varphi : G \to H$ is an injective group homomorphism. Prove that if K is a subgroup of G, then the index of K in G is the same as the index of $\varphi(K)$ in $\varphi(G)$.

Proof. Since φ is injective, $|G| = |\varphi(G)|$ and $|K| = |\varphi(K)|$, so by Lagrange's theorem,

$$[G : K] = \frac{|G|}{|K|} = \frac{|\varphi(G)|}{|\varphi(K)|} = [\varphi(G) : \varphi(K)].$$

□

5. Let M and N be normal subgroups of a group G such that $G = MN$. Prove that $G/(M \cap N) \cong (G/M) \times (G/N)$.

Proof. Define $\varphi : G \to (G/M) \times (G/N)$ by $g \mapsto (gM, gN)$. Then given $g, h \in G$,

$$\varphi(gh) = (ghM, ghN) = (gM, gN)(hM, hN) = \varphi(g)\varphi(h),$$

so φ is a homomorphism.

Now let $m \in M$ and $n \in N$. Then since N is normal, there exists $n' \in N$ such that

$$mn'm^{-1} = n \implies mn' = nm.$$

Then

$$\begin{aligned}\varphi(mn') &= (mn'M, mn'N) \\ &= (nmM, mn'N) \\ &= (nM, mN).\end{aligned}$$

Thus φ is surjective since all cosets of M (resp. N) have representatives in N (resp. M) because $G = MN = NM$.

Also, clearly $\ker \varphi = M \cap N$. By the first isomorphism theorem, then, $G/(M \cap N) \cong (G/M) \times (G/N)$. □

6. Let $G = S^1$, the unit circle in $\mathbb{C}$ under multiplication, and let $n \geq 2$ be an integer. Prove that the map $\varphi : G \to G$ given by $\varphi(x) = x^n$ is a surjective group homomorphism and identify its kernel. Conclude that G is isomorphic to a proper quotient of itself.

Proof. Let $g, h \in G$, and write $g = e^{ix}$ and $h = e^{iy}$. Then

$$\varphi(gh) = \varphi(e^{i(x+y)}) = e^{in(x+y)} = e^{inx}e^{iny} = \varphi(g)\varphi(h),$$

so φ is a homomorphism. Also, given any $e^{iz} \in G$, $\varphi(e^{iz/n}) = e^{iz}$, so φ is surjective. The kernel of φ is $\left\{e^{i(2\pi m)/n}\right\}$ where $0 \leq m < n$, which is nontrivial since $n \geq 2$. Thus the first isomorphism theorem lets us conclude that G is isomorphic to a proper quotient of itself. $\square$

7. Let C be a category. An object I of C is called *initial* if given any object X of C, there is a unique morphism $I \to X$ in C. There is a similar notion of *final* object of a category.

(a) Prove that initial objects in a category are unique up to unique isomorphism.

(b) Prove that the trivial group is both initial and final in the category of groups.

(c) Let G be a group and let N be a normal subgroup of G. Let C be the category whose objects are group homomorphisms $\varphi : G \to H$, such that $N \subset \ker \varphi$. A morphism from $\varphi_1 : G \to H_1$ to $\varphi_2 : G \to H_2$ in C is a group homomorphism $\psi : H_1 \to H_2$ such that $\varphi_2 = \psi\varphi_1$. Show that C is a category, and explain why the projection map $\pi : G \to G/N$ is initial in C.

Proof.

(a) Let I_1, I_2 be initial in a category C. Then there exist unique isomorphisms $\varphi : I_1 \to I_2$ and $\psi : I_2 \to I_1$, but in fact $\varphi^{-1} : I_2 \to I_1$ and $\psi^{-1} : I_1 \to I_2$ are also isomorphisms since the inverse of an isomorphism is an isomorphism, so uniqueness gives us $\varphi = \psi^{-1}$ and $\psi = \varphi^{-1}$, and we conclude that I_1 and I_2 are isomorphic up to unique isomorphism.

(b) Since all homomorphisms take the identity to the identity, the only option for a homomorphism from the trivial group to a given group G is the map φ sending the identity to the identity, and this is in fact a homomorphism since

$$\varphi(e \cdot e) = \varphi(e) = e_G = e_G \cdot e_G = \varphi(e) \cdot \varphi(e).$$

On the other hand, the only possible map from G to the trivial group is the one that sends everything to the identity, i.e. the trivial homomorphism.

(c) The identity morphism criterion is satisfied by the identity homomorphism on each object. Also, given objects $f : G \to H_1$, $g : G \to H_2$, and $h : G \to H_3$ along with morphisms $\varphi : H_1 \to H_2$ and $\psi : H_2 \to H_3$ such that $\varphi f = g$ and $\psi g = h$, we have $(\psi\varphi)f = \psi g = h$, so composition holds, and clearly the identity morphisms act as identities with respect to composition.

The statement that the projection map is initial in C is merely a restatement of the universal property of quotient groups (or quotient maps, if you prefer) in categorical terms. In fact, the fact that it is initial or final in some category is what justifies the name "universal property."

$\square$

8. Let G be a group, let H be a normal subgroup, let $x \in H$, and let $\mathcal{K} = [x]_G$. Our goal is to prove that $\mathcal{K}$ is the union of $[G : HC_G(x)]$ many conjugacy classes of H, all of equal size.

(a) Let $g \in G$ and $a \in H$. Show that $g \cdot ([a]_H) = \{ghg^{-1} : h \in [a]_H\}$ is equal to $[b]_H$ for some $b \in H$. Note that this gives a G-action on the set of all conjugacy classes of H.

(b) Suppose $\mathcal{K}$ is a conjugacy class in G, and that $\mathcal{K} \subset H$. Explain why H acts on $\mathcal{K}$ by conjugation, so that $\mathcal{K}$ is a union of H-conjugacy classes. Show that if $[a]_H$ is one such H-conjugacy class, then $g \cdot ([a]_H)$ is as well.

(c) Show that G acts transitively on the set of H-conjugacy classes contained in $\mathcal{K}$. Also, provide an explicit bijection between $[a]_H$ and $[b]_H$ for $a, b \in \mathcal{K}$.

(d) We now compute the stabilizer of an H-conjugacy class. To begin, show that if $g \in \mathrm{Stab}_G([x]_H)$, then $gxg^{-1} \in [x]_H$. Use this to conclude that $\mathrm{Stab}([x]_H) \subset HC_G(x)$. Then use Problem 1 to conclude the reverse containment.

(e) Use the Orbit-Stabilizer theorem (Problem 8 on Homework 3) to conclude that the number of H-conjugacy classes that union to give $\mathcal{K}$ is equal to $[G : HC_G(x)]$.

(f) For an application, take $G = S_n$ and $H = A_n$. Show that every S_n-conjugacy class of S_n contained in A_n is either an A_n-conjugacy class, or the union of two A_n-conjugacy classes of the same size. Prove that in this case, $[x]_{S_n} = [x]_{A_n}$ if and only if x commutes with some odd permutation.

Proof. NOT STARTED $\square$